From Hints to Textbooks

Studying Scaffolds through LLMs and Bayesian Methods

Justin Stevens

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Roadmap

1. The dataset and the two taxonomies
2. Method and infrastructure
3. Headline 1: counting & probability is different
4. Headline 2: harder problems have different transition matrices
5. My textbook vs. MATH number theory
6. Working with LLMs on the book
7. Takeaways

The dataset: SocraticMath

Where it comes from

- Competition problems from the MATH benchmark[a], each augmented with a *chain of Socratic questions*[b]
- One JSON file per problem: `problem`, `level` (1–5), `solution`, `type`, `socratic_questions`

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- 7 domains $\times$ 565–590 problems $=$ **4,043 problems**
- **31,613 Socratic hints** ($\approx$ 7.8 per problem)

[a] Hendrycks et al., *Measuring Mathematical Problem Solving with the MATH Dataset*, NeurIPS 2021.
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domain	problems
algebra	576
counting & probability	584
geometry	575
intermediate algebra	565
number theory	583
prealgebra	590
precalculus	570
total	4,043

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Taxonomy 1: four hint types

Goldin, Koedinger & Aleven (2013) — plus one extra type for Socratic chains:

[f] feature-pointing draws attention to a specific aspect of *this* problem
"Notice that these two angles are vertical angles"

[r] principle-stating supplies a general rule, definition, or theorem
"Vertical angles are equal in measure"

[b] bottom-out directly advances the solution — set up, substitute, solve
"Add these two known angles to find the missing one"

[e] extension pushes past the current problem — applications, generalizations
"Can you think of other situations where this matters?"

Goal: learn a 4×4 **transition matrix** M per group, where p_{ij} is the probability the next hint is type j given the current one is type i.

A worked example (algebra `hint_0`)

Let $f(x) = \begin{cases} ax + 3 & x > 2 \\ x - 5 & -2 \leq x \leq 2 \\ 2x - b & x < -2 \end{cases}$ Find $a + b$ if f is continuous.

1. [f] Can you explain what a piecewise function is and how it is represented algebraically?
2. [r] How would you determine if the function is continuous where different pieces meet?
3. [f] What conditions need to be satisfied for a piecewise function to be continuous?
4. [b] Can you identify the value of a and b that would make the function continuous?
5. [b] How would you approach solving for a and b?
6. [b] Are there any restrictions or constraints on the values of a and b?
7. [e] Can you explain the significance of continuity in real-life applications?
8. [e] Can you think of other examples where continuity is important?

Labels shown are my hand labels. On this chain Claude reproduces all 8/8

Taxonomy 2: Quarfoot's nine problem types

A musical metaphor, ordered (roughly) least $\rightarrow$ most mathematical substance:

	type	what it is
FN	First Notes	one-step, single-path, algorithmic; fluency
AC	Accidentals	looks routine, hides a twist; targets a misconception
CH	Chords	one idea, several steps; deepens a single concept
ET	Etudes	drills one technique; word-problem dressing
IM	Improvisations	a playground; first attempts fail, so you experiment
IN	Interpretations	wide-open solution space; compare qualitatively different paths
FP	First Pieces	first problems needing multiple ideas plus a macro-level plan
SP	Showpieces	very hard, hinges on a flash of insight; cognitive
MP	Masterpieces	many skills converging on a generalizable truth; capstones

The nine types, illustrated from my number theory textbook

Claude's classification of *Olympiad Number Theory* problems (533 total), one example each:

type	problem	source
FN	Find x such that $1 + x + x^2 + x^3 + \cdots = 4$.	Mandelbrot
AC	Show there are no integers x, y with $1691x + 1349y = 1$.	Ch. 5
CH	The sum of 18 consecutive positive integers is a perfect square. Find the smallest possible such sum.	AMC 12
ET	Compute $\sum_{m=1}^{75}(2 + 3m)$.	Ch. 1
IM	Pages numbered 1–n; one page number was accidentally added twice, giving 1986. Which page?	AIME
IN	Prove these Fibonacci identities *in two different ways*.	Ch. 2
FP	Find the 8th term of $1440, 1716, 1848, \ldots$, formed by multiplying corresponding terms of two arithmetic sequences.	AIME
SP	Evaluate $\prod_{n=2}^{\infty} \frac{n^3-1}{n^3+1}$.	Putnam
MP	Prove every positive integer is a sum of non-consecutive Fibonacci numbers.	Zeckendorf

Method: one Claude call per problem, two classifications

- `claude-opus-4-8` receives the problem, difficulty level, worked solution, and the numbered Socratic chain, and returns strict JSON:
  1. one hint label (`f`/`r`/`b`/`e`) per question, in order
  2. one Quarfoot type for the whole problem
- Prompt contains both taxonomies' definitions + the worked example above as the single few-shot anchor
- Every response is **validated** (label count = question count, legal codes only) and retried on failure — batch retry, then per-problem retry
- Downstream analysis: group the chains (by domain, or by problem type), count transitions, Dirichlet–multinomial posterior per group, row-averaged symmetrized KL, similarity e^{-KL}
- Artifacts: one JSON per problem in `classifications_llms/`, figures in `outputs_llms/`

Infrastructure: written by Claude, run on Claude

Built with Claude Code

- Agent wrote the backend, classifier, CLI commands, and plots inside the existing `uv` project
- Backend auto-selects: Anthropic SDK if an API key is set, else headless `claude -p` calls on my Claude subscription
- Batched 10 problems/call ($\approx$ 410 calls), 12 parallel workers

Robustness (needed it!)

- Resumable: each problem's JSON written on arrival; re-runs skip finished work
- Circuit breaker: 3 consecutive all-fail batches $\Rightarrow$ stop (usage limit), auto-resume later

Compute & cost

- Throughput $\approx$ 53 problems/min $\Rightarrow$ $\approx$ **80 min of active compute** for all 4,043 problems ($\approx$ 3 h wall clock: hit the subscription usage limit twice, auto-resumed)
- Marginal cost: **$0** on the subscription ($\approx$ $60 API-equivalent at Opus list prices)

It starts with staring at the raw transition matrices

Per-domain 4×4 matrices over [f][r][b][e]. Six of the seven look roughly like algebra's. One doesn't:

$$M_{\text{algebra}} = \begin{pmatrix} \mathbf{0.227} & 0.224 & 0.498 & 0.051 \\ 0.160 & 0.272 & 0.502 & 0.067 \\ 0.089 & 0.158 & 0.600 & 0.153 \\ 0.030 & 0.073 & \mathbf{0.193} & \mathbf{0.704} \end{pmatrix} \quad M_{\text{count. \& prob.}} = \begin{pmatrix} \mathbf{0.418} & 0.126 & 0.421 & 0.035 \\ 0.184 & 0.242 & 0.441 & 0.133 \\ 0.070 & 0.104 & 0.715 & 0.111 \\ 0.030 & 0.060 & \mathbf{0.079} & \mathbf{0.831} \end{pmatrix}$$

- $f \to f = 0.418$: the highest of *any* domain (runner-up: prealgebra 0.331) and nearly **2**$\times$ algebra's 0.227 — c&p hints chain observation $\to$ observation
- $e \to b = 0.079$: the lowest of any domain — once a c&p chain reaches extensions it almost never drops back into computation ($e \to e = 0.831$)
- The other rows are unremarkable — the signal is concentrated, not diffuse

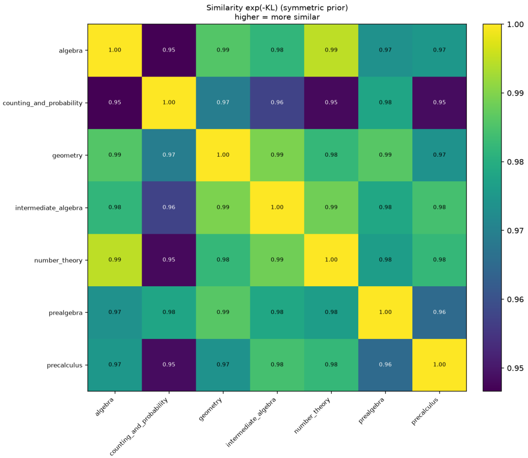
Compressing a comparison into one number

Count transitions, add the all-ones Dirichlet prior, normalize rows — e.g. the counting & probability f-row has counts $(491, 147, 495, 40)$, so $p_{ff} = \frac{491+1}{1173+4} \approx 0.418$. Then compare two domains row by row: symmetrized KL, averaged over the four rows.

row	f	r	b	e	mean
$\frac{1}{2}[KL(A\|C) + KL(C\|A)]$	**0.096**	0.030	0.030	0.063	**0.0549**

Similarity $= e^{-0.0549} \approx 0.947$ — the *lowest* of any domain pair. The f-row does the most work, exactly the observation-chaining row from the previous slide.

Across all 21 pairs: counting & probability is the outlier



- The dark stripe is counting & probability's row/column: mean similarity to the other domains **0.958**, the lowest of any domain
- Geometry — the domain you might have guessed — is among the most *central* (mean 0.981)
- One-vs-rest permutation test: c&p diverges most from the pooled rest — sym. KL 0.034, $\approx 1.7\times$ the runner-up; Holm-adjusted $p < 0.001$

Why counting & probability reads differently

- Its hint *marginals* already stand out — most feature-pointing, least principle-stating of any domain:

	f	r	b	e
counting & probability	**23.9%**	**13.9%**	50.0%	12.1%
all domains (mean)	18.9%	18.6%	51.5%	10.9%

- And its *dynamics*: hints stack observations ($f \to f$ at 0.418) before computing, and once a chain reaches extensions it stays there ($e \to e$ at 0.831 vs. 0.704 for algebra)
- Intuition: "notice how many ways..." beats "recall the theorem..." — combinatorics hints scaffold by *observation*, not by *principle*

Takeaway: the outlier domain isn't the one with the strangest notation — it's the one whose scaffolding *strategy* differs.

Split the substance axis in half: routine vs. complex

Group the same 4,043 chains by Claude's *Quarfoot type*, split at the axis midpoint: **routine** = FN/AC/CH/ET (2,880 problems, 19,046 transitions) vs. **complex** = IM/IN/FP/SP/MP (1,163 problems, 8,524 transitions):

$$M_{\text{routine}} = \begin{pmatrix} 0.283 & 0.219 & 0.452 & 0.046 \\ 0.134 & 0.303 & 0.482 & 0.081 \\ 0.079 & 0.142 & \mathbf{0.628} & \mathbf{0.150} \\ 0.017 & 0.058 & 0.136 & 0.789 \end{pmatrix} \quad M_{\text{complex}} = \begin{pmatrix} 0.338 & 0.156 & 0.486 & 0.020 \\ 0.177 & 0.270 & 0.525 & 0.029 \\ 0.071 & 0.080 & \mathbf{0.771} & \mathbf{0.078} \\ 0.064 & 0.046 & 0.145 & 0.746 \end{pmatrix}$$

- $b \to b$: $0.628 \to \mathbf{0.771}$ — complex problems' hints run long *bottom-out constructions*; the b-row alone carries the most KL
- The whole e-column collapses: $b \to e$ halves ($0.150 \to 0.078$), and extensions fall from 5.9% of hints to **2.0%**
- Principle-stating share drops $23.9\% \to 15.1\%$ — fewer "recall the theorem" moves

Similarity $e^{-0.0386} = \mathbf{0.962}$; problem-level permutation test: $p < 10^{-4}$ (0 of 10,000 shuffles reach it). Only **4 of 21** domain pairs are farther apart — all four involve counting & probability.

What "building" looks like (c&p `hint_157`, First Pieces)

Four standard six-sided dice are rolled. If the product of their values is even, what is the probability their sum is odd?

Level 5 — 1 [f], 1 [r], **13** [b], 0 [e]

1. [f] Can you explain the problem statement in your own words?
2. [r] What is the condition for the product of the dice values to be even?
3. [b] If all the dice rolls yield odd numbers, how many possible outcomes are there?
4. [b] How many possible outcomes are there in total?
5. [b] How can we determine the number of outcomes where at least one even number is rolled?

⋮ seven more [b]: split into cases by how many dice are odd, count each case, combine

13. [b] What is the desired probability of the sum being odd?
14. [b] Can you simplify the fraction to its lowest terms?
15. [b] What is your final answer for the probability?

- Each hint lays one brick — complement, case split, count, combine, simplify — a *construction schedule* for the argument; when the object is built, the chain just ends (no [e] coda)
- The double edge: bottom-out is the tutoring literature's "just tell me" hint — the chain hands over the macro-plan pre-chopped into executable steps, the very planning skill First Pieces exist to train

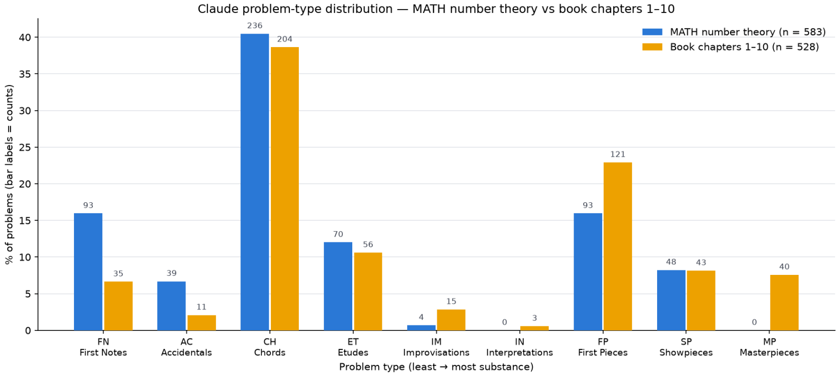
Not a threshold but a gradient — and where it ends

type	n	r share	$b \to b$	e share
FN	658	31%	0.56	10%
AC	289	23%	0.63	5%
CH	1,501	24%	0.63	5%
ET	432	16%	0.69	4%
IM	13	16%	0.64	4%
FP	661	16%	0.77	2%
SP	487	13%	0.78	2%
MP	2	10%	—	0%

IN: 0 problems; MP: 2 — too few to estimate a matrix.

- Not an artifact of where the line is drawn: moving FN $\to$ SP, principle-stating falls 31% $\to$ 13%, $b \to b$ climbs 0.56 $\to$ 0.78, extensions fade 10% $\to$ 2%
- Routine problems are *vehicles for a rule*: hints state it [r], apply it, then generalize away from the problem — the [e] coda
- Complex problems' difficulty is *architectural* — no single rule to recall — so the hints enumerate the plan instead: **hints stop teaching and start building**
- The capstone end of the axis is nearly absent here: 2 Masterpieces and 0 Interpretations in all of MATH. Where do capstones actually live?

Same classifier, two corpora: MATH number theory vs. book ch. 1–10



- The *middles* match: Chords 40.5% vs. 38.6%; Etudes and Showpieces within ∼1.5 points
- The *tails* diverge in opposite directions

Reading the tails as pedagogy

MATH number theory ($n = 583$)

- Thick floor of fluency checks: FN + AC = **22.6%** ("what is the units digit of…")
- Zero Masterpieces; only 25% of problems in the complex half of the axis
- A competition *item bank*

Book chapters 1–10 ($n = 528$)

- Light on drills: FN + AC = 8.7%
- **42%** in the complex half: FP 22.9% vs. 16.0%
- 7.6% Masterpieces — culminating capstones (Zeckendorf, USAMO, Putnam)
- A curated *learning arc*

The arc is visible *within* the book too: the share of high-substance problems (FP/SP/MP) climbs from **24%** in Ch. 1 to ≈ **50%** by Ch. 9–10.

Why this is credible: identical classifier, prompt, and rubric on both corpora — near-identical middles, sharply different tails ⇒ properties of the corpora, not classifier noise.

What this offers people who write textbooks

- **An audit tool for your problem mix.** Type profiles make the shape of a book quantitative: drills vs. concept-deepeners vs. capstones, per chapter and overall — and comparable against item banks or other texts.
- **A check on the arc.** You can verify the substance axis actually climbs the way you intended (mine does: $24\% \to 50\%$ high-substance).
- **Gap-finding.** What I learned about my own book:
  - essentially no *Interpretations* (3 of 533) — almost nowhere do I ask students to solve one problem *two different ways* and compare
  - Improvisations are rare too (15) — few low-stakes "playground" problems
  - the capstones I hoped were there really do register (40 Masterpieces)
- **For tutoring systems:** hint scaffolding should adapt per domain — combinatorics wants observation-chaining, algebra wants principle-then-compute.

Working with LLMs on the book: observations

Working with LLMs on the book: observations (cont.)

Takeaways, caveats, next steps

Takeaways

1. Counting & probability scaffolds by *observation*, not principle — the clear domain outlier, with the mechanism visible in the raw matrices
2. Complexity shows up in the dynamics: complex problems' chains run long bottom-out constructions and drop principle-stating and extensions — hints stop teaching and start building ($p < 10^{-4}$)
3. Problem-type profiles act as corpus fingerprints: item bank vs. curated arc — and the whole pipeline is cheap enough to be routine ($\approx$ 80 min, \$0 marginal)

Caveats

- No human-labeled evaluation set yet — the earlier "gold set" was itself LLM-authored, so I dropped it; a real one is the top priority
- The capstone end of the axis is nearly absent from MATH (2 Masterpieces, 0 Interpretations) — the complex half is effectively IM/FP/SP
- Claude saw the worked solution: single-solution write-ups may bias it against open-ended types (IM/IN) and Masterpieces; one model, one prompt — no inter-rater κ yet

Next steps: human gold set + Cohen's κ with a second annotator; generate Socratic chains for the book's complex-half problems (including its 40 Masterpieces) to test headline 2 where capstones actually live; hint transitions vs. difficulty level; HMM over noisy labels.

Thanks! Questions?

Code available here.